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Fortifying Deepfake Detection With AI and Postquantum Cryptography-Based Watermarking Technique

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Abstract

To future-proof digital media authentication against quantum computing decryption threats, this article proposes a hybrid system integrating lattice-based post-quantum cryptography (Kyber/Dilithium signatures) with a deep neural audio watermarking scheme.

Key Methodology & Architecture

Lattice-based Post-Quantum Cryptography (PQC), DWT-LSB embedding, Deep Binary Neural Network (BNN) detection, SHA3 hash integrity verification.

Datasets & Benchmark Environments

VoxCeleb1, CREMA-D, Custom Broadcast News Audio Corpus.

Performance Metrics & Graph Axis Parameters

Reported Results: Security Level = 128-bit Post-Quantum, Detection Accuracy = 99.05%, Latency = 12ms per audio second.

Graph Parameter Mapping: X-axis: Cryptographic Key Length (Bits: 512, 1024, 2048, 4096); Y-axis: Signature Generation & Embedding Time (ms).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Quantum-resistant cryptographic integrity verification coupled with AI watermarking.	Larger signature payload sizes slightly reduce payload bandwidth.

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